Why Risk it, When You Can {rix} it: A Tutorial for Computational Reproducibility Focused on Simulation Studies

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Abstract

Computational reproducibility remains limited in psychological research, despite widespread norms for sharing data and analysis code. One reason is that reproducibility exists on a continuum, ranging from partial transparency—such as providing scripts or software version numbers—to fully executable research compendia that regenerate all results from raw code. In this article, we introduce Nix and the {rix} R package as a practical framework for achieving full computational reproducibility in simulation-based research. We provide a step-by-step tutorial demonstrating how {rix} can be used to define, build, and share isolated, project-specific software environments that precisely capture R versions, package dependencies, system libraries, and integrated development environments. We further illustrate this workflow by reproducing a complete manuscript using Quarto and the {apaquarto} extension, showing how analyses, figures, and text can be regenerated in a single, executable pipeline. Together, these tools lower the technical barrier to robust, end-to-end reproducibility and offer a scalable solution for simulation studies and methodological research in psychology and related fields.

Keywords: reproducibility, Nix, simulation studies, R, computational methods

Why Risk it, When You Can {rix} it: A Tutorial for Computational Reproducibility Focused on Simulation Studies

Psychological science is in the midst of a credibility revolution (Vazire, 2018), yet much of that scrutiny has focused on empirical studies. Monte Carlo simulation studies (hereafter referred to as simulation studies) have received far less (Siepe et al., 2024), even though they are a primary tool for evaluating the statistical methods (Morris et al., 2019) and may suffer from selective reporting (Bouma et al., 2026; Pawel et al., 2024, 2025). One aspect of credibility that remains especially underexamined for simulations is reproducibility (i.e., the ability to reproduce the results of a study based on provided data, code, materials, and software/hardware) (Hardwicke et al., 2020). In a simulation study this involves more than reanalyzing a fixed dataset, because the code also generates the data. Reproducing such a study therefore entails reproducing the data generation, the analysis, and the summary of results. Montoya and Anderson (2026) conceptually address each of these stages separately.

Reproducibility in this sense is not all-or-nothing but occupies a continuum (Peng, 2011), on which a study is positioned according to the extent of the materials that are publicly available. This ranges from publication alone at the low end to code and data that are linked and directly executable at the high end. Reaching that executable end is often treated as *full* reproducibility (Peikert et al., 2021; Peikert & Brandmaier, 2021), and open science initiatives have pushed work toward it (Kidwell et al., 2016; Levenstein & Lyle, 2018; Nosek et al., 2015). Yet code and data are never fully self-sufficient (Epskamp, 2019; Wiebels & Moreau, 2021; Ziemann et al., 2023). Even error-free code depends on a hierarchy of software components, including the programming language version, the packages used in the analysis, and the system libraries those packages rely on. We define these and related computing terms, which may be unfamiliar to readers without a computer science background, in Table 2. When these differ from the originals, code may fail, behave inconsistently across machines, or yield conflicting numerical results (Baker et al., 2024; Glatard et al., 2015; Hodges et al., 2023; Nosek et al., 2022). These problems are particularly acute for simulation studies, which typically rely on a large amount of interconnected code, packages versions, and the underlying system itself to generate their data and produce the results of interest (Luijken et al., 2024; Siepe et al., 2024).

To make this concrete, we use *computational environment* to refer to the complete software context required for an analysis to run successfully: the programming language version, package versions, system libraries, and operating system (Figure 1) (Rodrigues, 2023; Rodrigues & Baumann, 2026). We define *computational environment reproducibility* as the ability to reconstruct this entire set of software dependencies on any machine and at any future time, such that executing the same code yields the same numerical results[1]. Empirical assessments show that current practice falls short of this ideal. Indeed, Siepe et al. (2024) report that, among simulation studies published in psychology in 2021 and 2022, nearly two-thirds provide no accompanying code, and among those that do, documentation of the computational environment is rarely included. This gap is consequential, as simulation studies inform methodological recommendations, meaning that insufficient reproducibility may undermine confidence in those recommendations (Luijken et al., 2024; White et al., 2024).

These challenges may persist for several reasons. Arguably, one is that even researchers aware of these demands face a fragmented landscape of solutions, each addressing only part of the problem. Package-level managers such as {renv} (Ushey, 2024) and {groundhog} (Simonsohn, 2020) stabilize R package versions but do not manage the R interpreter itself or the system-level libraries those packages depend on (see Table 2). Workflow orchestration tools such as {targets} (Landau, 2021) and Make (Feldman, 1979) support reproducibility in a different sense: they specify the structure of an analysis by formalizing the order in which steps should run and by tracking dependencies among intermediate results. These tools clarify *how* an analysis proceeds, but they assume that the software stack required to run each step is already stable. Containerization tools such as Docker, including R-focused implementations like Rocker (Boettiger, 2015; Boettiger & Eddelbuettel, 2017) offer a more comprehensive approach by

[1] We note that the definition of computational environment reproducibility in this context may be considered rather loose. For the interested reader, we refer to Malka et al. (2026) (p. 3) and Lamb and Zacchiroli (2022).

bundling the full environment (i.e., operating system, system libraries, interpreter versions, and packages) into a single executable image. Yet their use requires familiarity with Linux system administration, and even containerization may suffer from temporal drift when Dockerfiles rely on mutable upstream repositories (Malka et al., 2026). For a detailed comparison of these tools and their limitations, see Rodrigues and Baumann (2026). Researchers thus face a difficult choice between solutions that are accessible but incomplete or approaches that are powerful but demand technical expertise.

In this article, therefore, we focus specifically on computational environment reproducibility as the foundation upon which other reproducibility practices depend. For that, we introduce Nix (Dolstra et al., 2004), a functional software[2] ecosystem designed to make software installation deterministic, and {rix} (Rodrigues & Baumann, 2025), an R interface that allows researchers to use Nix without needing deep knowledge of its underlying language or infrastructure. Our main objective with the tutorial is not to introduce a specific workflow orchestration system or to prescribe a particular analytic structure.[3] Instead, we aim to show what Nix and {rix} are and how they can establish a stable, cross-platform environment within which any simulation study—whether organized in simple, documented script sequences (e.g., .R files that `source()` others), through more formal orchestration tools (e.g., {targets}) or embedded as code chunks in `.Rmd` or `.qmd`—can be executed reliably.

We illustrate these ideas through a reproducible simulation study conducted in R, culminating in this automated APA-formatted manuscript generated with `apaquarto` (Schneider, 2024). Although the example centers on R because of its prominence, the principles underlying environment reproducibility with {rix} apply equally to other languages, including Python and

[2] *Functional* is used in the sense of functional programming (pure functions, immutability, explicit inputs), not in the colloquial sense of *working*.

[3] We leave workflow orchestration aside for two reasons. First, many tools address it, each with its own conventions, and doing them justice would require a more tailored treatment than a single tutorial allows. Second, although formal orchestration may be necessary for full reproducibility, it rests on a stable computational environment: orchestrating analyses is only meaningful when the software they run on is itself reproducible. A study whose scripts are well documented, whose file dependencies are explained, and whose environment is stable therefore already sits high on the reproducibility spectrum, and that foundation is what we focus on here. For readers interested in the workflow side, we suggest Peikert et al. (2021) and van Lissa et al. (2021).

Julia, and to different development environments such as RStudio, VS Code, Emacs, or Positron. At the end of the article, we also briefly introduce {rixpress} (Rodrigues, 2025), a workflow orchestration tool that builds on Nix to coordinate pipelines across R, Python, and Julia.

Setting the Stage: A Simulation Study and Its Reproducibility Risks

Imagine you have just been awarded a grant to conduct a large-scale simulation study. The study is designed to evaluate the performance of a statistical estimator under varying data-generating conditions (see Appendix A for full technical details). This tutorial is organized around this scenario. We use this example to ground our discussion into a typical methods section, but readers can follow the tutorial without engaging deeply with the simulation itself. In practice, simulation studies are typically organized into multiple component files. To mimic this practice, our simulation is organized into five sequential scripts: data generation (`01_data_generation.R`), model specification (`02_models.R`), simulation execution (`03_run_simulation.R`), performance metric calculation (`04_performance_metrics.R`), and results visualization (`05_plots.R`) [4].

Now suppose a researcher attempts to reproduce the simulation run and results reported in the article. What might prevent them from obtaining identical outcomes, assuming error-free code? The natural first concern is the version of packages. Installing R packages at a later time may lead to errors if functions have been renamed or deprecated (e.g., `dplyr::cur_data()` changed to `pick()`, or, although usually discouraged, internal functions, such as `lavaan:::lav_utils_get_ancestors`, renamed to `lavaan:::lav_graph_get_ancestors`), or to subtly different results due to changes in default settings (e.g., `stringsAsFactors` defaulting to FALSE as of R 4.0). Beyond package versioning, many packages rely on system-level libraries that must be installed separately from R. Our simulation illustrates this dependency structure directly: the {rvinecopulib} package interfaces with a C++ backend and

[4] We note that, for the sake of simplicity, we included all code directly within this document as executable chunks in a single `.qmd` file. This entails that the simulation runs from start to finish (i.e., generating data, producing results and figures automatically. This approach would be impractical for many real-world simulation studies, which are often too computationally intensive. We return to this trade-off later in the tutorial as the solution is quite simple: one may simply run the simulation outside the `.qmd` file and load the results for the analyses in the same file.

links against external libraries such as Boost, Eigen, and RcppThread (Nagler & Vatter, 2025).

The R language version introduces another layer of dependency. Code written for R 4.0 may rely on syntax or functionality that is unavailable in earlier versions (e.g., the native pipe `|>` introduced in R 4.1). More subtly, changes to R's random number generation across major versions mean that identical code executed with the same seed can nevertheless produce different random sequences (Ottoboni & Stark, 2018). This is especially consequential for simulation studies, because the random draws are not incidental to the analysis but constitute the data itself. Indeed, a simulation study regenerates its data throughout the multiple runs. The random number generator is therefore part of the computational environment: reproducing the data requires not only the same seed but the same generator algorithm (e.g., Mersenne-Twister) and R version. Moreover, reproducing every estimate exactly can depend on lower-level components still, such as the numerical libraries R uses for matrix computations (e.g., BLAS/LAPACK). With respect to the latter, we do not claim that this exact level of identity must be met. How much reproducibility is required is ultimately for the scientific community to decide (e.g., through journals and disciplinary norms), and reproducing the broader pattern of results is arguably enough. Regardless, the solution we propose increases the chances of reaching such a level of reproducibility, consistent with empirical findings (Malka et al., 2026).

Finally, analyses embedded in a literate programming workflow (documents that combine narrative text and executable code) add further dependencies. Rendering R Markdown (`.rmd`) or Quarto (`.qmd`) documents, for example, requires a document conversion tool (such as Pandoc) and a typesetting system (such as LaTeX), each with its own version and installation requirements. Reproducibility therefore depends not only on code and data, but on the broader computational environment in which analyses run.

Nix and {rix}: A Comprehensive Solution

A potential solution to this problem is Nix (Dolstra et al., 2004), a software ecosystem whose primary concern is reproducible, declarative builds. To achieve this Nix is a programming language, a build tool and a package manager. This article focuses mostly on Nix the package manager. Most package managers (think of Apple's or Android's app stores) are imperative: they modify a system's state as they install or update software. Nix, in contrast, treats build instructions and dependencies as immutable, enforcing reproducible, declarative, and isolated environments across platforms. This allows researchers to specify exactly which versions of programming languages and packages an analysis requires, while Nix automatically resolves and pins the underlying system libraries, so the environment can be recreated reliably on any machine.

Core Principles

Rather than installing software into global directories (e.g., `/usr/lib`), Nix places every package in its own directory under `/nix/store`. Each package path contains a cryptographic hash representing its precise inputs: source code, dependencies, and build instructions. Because these paths are content-addressed, multiple versions of the same software can coexist without conflict. A researcher can, for example, maintain projects requiring R 4.1.0 and R 4.3.3 side by side, or use different package versions across analyses (Rodrigues & Baumann, 2025).

The Nix ecosystem is built around nixpkgs, a version-controlled repository comprising more than 120,000 packages, including nearly all of CRAN and Bioconductor. By pinning a specific commit or date, researchers freeze the entire software stack (i.e., R itself, R packages, and all system libraries). This eliminates the system-dependency problems that tools like {renv} cannot address (Rodrigues & Baumann, 2025). Empirical work has shown strong rebuildability and reproducibility rates for historical nixpkgs snapshots (Malka et al., 2026; Rodrigues & Baumann, 2026), and binary caches, which often allow environments to materialize in seconds, make Nix practical for interactive research workflows (Rodrigues & Baumann, 2025).

The {rix} Package: R Interface to Nix

As previously stated, Nix is also a functional programming language. Because the Nix package manager is declarative, it requires expressions written in this language to install software. However, since this language is unfamiliar to most researchers, we recommend and focus on using {rix} to lower this barrier. The {rix} package provides an R-native interface: a single call to `rix()` generates complete Nix configurations from standard R syntax, specifying R versions, CRAN packages, system libraries, and even Python or Julia components when required. Users never need to read or write Nix code directly, as {rix} translates automatically. For more on {rix}, including its integration with rstats-on-nix for dated, reproducible snapshots, see Rodrigues and Baumann (2025).

We also note that, although Nix is capable of replacing tools like Docker for isolation or {renv} for package management, it does not require an all-or-nothing transition. Researchers can adopt it gradually and use it alongside familiar tooling. For instance, by building Docker images with Nix, converting existing {renv} lockfiles, or running {targets} pipelines within a Nix-defined environment (Rodrigues & Baumann, 2025). This allows Nix to strengthen reproducibility while preserving established workflows. We will come back to this after the tutorial.

A Practical Example: Setting up a Reproducible Simulation Study with {rix}

We take the perspective of a researcher beginning a simulation study who wants it reproducible from the outset. Rather than writing the analysis first and reproducing it later, we set up the computational environment with {rix} first, then create and run the scripts and manuscript inside it. We then show how another researcher could reproduce it.

Step 1: Installing Nix and {rix}

Before proceeding, both Nix and the {rix} R package need to be installed. The quick start below covers the common case, but the complete, up-to-date instructions for each operating system are maintained in the {rix} documentation: Linux and Windows (WSL2) (https://docs.ropensci.org/rix/articles/setting-up-linux-windows.html) and macOS (https://docs.ropensci.org/rix/articles/setting-up-macos.html).

The installation is the same across operating systems, with one prerequisite on Windows: Nix runs inside the Windows Subsystem for Linux 2 (WSL2), which must be installed and configured first (see the {rix} documentation above). On all systems, install Nix by running the Determinate Systems installer in a terminal[5]. To open it, on macOS, go to Applications →

[5] A terminal is a text-based way to give your computer instructions. Readers new to the command line may find the following link a useful primer: https://swcarpentry.github.io/shell-novice/. This was also suggested in Wiebels and

Utilities → Terminal. For Windows, launch your WSL2 (Ubuntu) shell from the Start menu. For Linux, open your distribution's terminal application. Then run (Listing 1):

Listing 1 Installing Nix

```
curl --proto '=https' --tlsv1.2 -sSf \
  -L https://install.determinate.systems/nix | \
  sh -s -- install
```

Once Nix is installed[6], there are two ways to access {rix}, depending on whether R is already installed on your system. In this tutorial, we proceed as if R was already installed[7], as that most likely will be the case for most readers (Listing 2):

Listing 2 Installing {rix} from CRAN or developmental version

```
# CRAN version
install.packages("rix")
# Development version
install.packages(
  "rix",
  repos = c(
    "https://ropensci.r-universe.dev"
  )
)
```

Step 2: Specifying the Computational Environment

We then begin with an empty folder for this project (here named `Why-risk-it-when-you-can-rix-it`). There are no simulation scripts or manuscript inside it yet. The only file we write by hand for now is `generate-env.R` (or similar), which uses the

Moreau (2021)

[6] It is worth noting that {rix} can generate Nix expressions even without Nix installed on your system: you can write a `default.nix` file without Nix, but you cannot build or enter the resulting environment unless Nix is installed (Rodrigues & Baumann, 2025).

[7] We, however, recommend uninstalling your local R and letting Nix manage R, R packages, and other tools entirely. This approach avoids potential conflicts between system-installed and Nix-managed software, an issue we will illustrate later in this tutorial. See https://docs.ropensci.org/rix/articles/setting-up-linux-windows.html#case-1-you-dont-have-r-installed-and-wish-to-install-it-using-nix-as-well for more details.

`rix()` function from the {rix} package to produce a `default.nix` file: a declarative specification of all software dependencies the project needs.

We start with the smallest possible environment: R version and an IDE. We add packages as the simulation takes form and extend the environment again for the manuscript later. This bare specification is shown in Listing 3 (the complete version, which covers the simulation and the manuscript, is on the GitHub repository, https://github.com/felipelfv/Why-risk-it-when-you-can-rix-it, as `gen-env.R`):

Listing 3 Bare environment specification: R and RStudio, no packages yet

```
library(rix)

# choose the path to your project
project_directory <- "." # here, the "Why-risk-it-when-you-can-rix-it" folder

rix(
  date = "2026-01-14", # recommendation: pin a date
  ide = "rstudio", # editor to include
  project_path = project_directory,
  overwrite = TRUE
)
```

Environment Specification

The `rix()` function[8] constructs the specification through a series of parameters that collectively describe the computational environment. Each parameter serves a distinct purpose in defining the environment's characteristics, and we will clarify those throughout this tutorial section.

Specifying the R version. Researchers must first determine which version of R to use. This can be accomplished in two ways: The `r_ver` argument accepts an exact version string (e.g., "4.3.3") or special designations such as "latest-upstream" for the most recent stable release. Alternatively, the `date` argument specifies a particular date (e.g., "2024-11-15"), which ensures

[8] For an overarching information on the function `rix()`, we suggest the following {rix} documentation: https://docs.ropensci.org/rix/articles/project-environments.html

that R and all packages correspond to the versions available on that date. The date-based approach is generally preferable for reproducibility, as it captures a complete snapshot of the R ecosystem at a single point in time. For this tutorial, as shown on top, we use the `date` parameter to ensure temporal consistency across all software components (Rodrigues & Baumann, 2025) (see {rix} documentation for more: https://docs.ropensci.org/rix/articles/project-environments.html).

Configuring the development environment. The `ide` parameter controls whether an integrated development environment (IDE) is included in the Nix environment, allowing users to interactively develop and run code within their editor of choice. When `ide` is specified, the project can be opened directly in the corresponding IDE, with all dependencies provided by the Nix environment. Most readers likely already have an IDE installed locally. Here we instead show how to open RStudio from the rix specification, which does not require RStudio to be installed on your system. For example, setting `ide = "rstudio"` installs a project-specific version of RStudio inside the Nix environment. This is required for RStudio because, unlike most other editors, it cannot attach to an external Nix shell unless it is itself installed via Nix. On macOS, RStudio is only available through Nix for R versions 4.4.3 or later (or environments dated 2025-02-28 or later). For earlier R versions, alternative editors must be used. Other supported IDEs include Positron (`ide = "positron"`), Visual Studio Code (`ide = "code"`), and command-line interfaces such as Radian (`ide = "radian"`). These tools may either be installed directly within the Nix environment using the `ide` parameter, or users may rely on an existing system installation by setting `ide = "none"` (or `ide = "other"`) and configuring `direnv` to automatically load the Nix environment when the project directory is opened[9]. All IDEs installed via Nix are project-specific and do not interfere with system-wide installations. Detailed configuration instructions are provided in the {rix} documentation:
https://docs.ropensci.org/rix/articles/configuring-ide.html

[9] **`direnv`** is a lightweight utility that integrates with the user's shell and automatically loads project-specific environment settings when navigating into a directory (via a `.envrc` file), and unloads them when leaving. This makes environment activation implicit and reduces the risk of running analyses in the wrong software context.

Setting file output parameters. The `project_path` parameter indicates where the `default.nix` file should be written ("." denotes the current directory), while `overwrite` controls whether an existing file should be replaced. Adding to this, setting `print = TRUE`, which is another argument, displays the generated specification in the console for immediate verification (Rodrigues & Baumann, 2025).

Step 3: Generating the Environment Specification

After defining the computational environment, the `rix()` function must be executed to generate the `default.nix` file. This can be done interactively by running `rix()` (Listing 3). The file is written to the folder given in `project_path`, so you can confirm it succeeded by checking that this folder now contains `default.nix`.

Step 4: Building and Using the Reproducible Environment

Once Step 3 is complete, we build the reproducible environment by navigating to the project directory in the terminal. In our case, this is the `Why-risk-it-when-you-can-rix-it` folder. From there, run the following command (Listing 4):

Listing 4 Building the Nix environment

```
# in the terminal, from the project directory:
nix-build
```

The expected output should look similar to (Listing 5):

Listing 5 Expected output from `nix-build`

```
unpacking 'https://github.com/rstats-on-nix/nixpkgs/archive/2025-08-25.tar.gz'
  into the Git cache...
warning: ignoring untrusted substituter...
warning: ignoring the client-specified setting...
/nix/store/qa7fq20m2f94szsnqzciwv8h4n81w43v-nix-shell
```

This command builds the environment according to the specification. The first execution will download and install all required packages, which may take a few minutes depending on network speed and system resources. Subsequent builds use cached packages and complete in seconds. Upon successful completion, a path to the constructed environment in the Nix store is printed (here, `/nix/store/qa7fq20m2f94szsnqzciwv8h4n81w43v-nix-shell`), and a symbolic link named `result` appears in the project directory pointing to this location. In our illustration thus far, no packages were added yet.

Note that the warnings indicate that you are not configured as a trusted user, so Nix cannot use the rstats-on-nix binary cache and will instead compile packages from source, which is slower. To enable binary caching, see the {rix} documentation: https://docs.ropensci.org/rix/articles/binary-cache.html.

To activate the environment, run (Listing 6):

Listing 6 Activating the Nix environment

```
# in the terminal, from the project directory:
nix-shell
```

The expected output (if you have configured yourself as a trusted user, otherwise the same warnings will appear) should look similar to (Listing 7):

Listing 7 Expected output from `nix-shell`

```
unpacking 'https://flakehub.com/f/DeterminateSystems/nixpkgs-weekly/...'
  into the Git cache...
[nix-shell:~/Why-risk-it-when-you-can-rix-it]$
```

This command drops the user into a shell where all specified packages and tools are available. The shell prompt changes to indicate that a Nix environment is active (here, `[nix-shell:~/Why-risk-it-when-you-can-rix-it]$`). To verify that R is being provided by Nix rather than a system installation, run `which R`. This should return a path within `/nix/store/`. Moreover, from within the Nix shell, users can launch their IDE by typing its name (e.g., `rstudio` or `positron`), which opens the IDE with the Nix environment active (Listing 8)[10]:

[10] Please note that activating an RStudio instance via Nix does **not** automatically open the specific project directory

Listing 8 Activating RStudio

```
[nix-shell:~/Why-risk-it-when-you-can-rix-it]$ rstudio
```

Step 5: Developing and Making the Simulation Reproducible

With RStudio open from the Nix environment, we write the five simulation `.R` scripts. For the sake of clarity, one does not need to use RStudio, but we illustrate it because most readers probably make use of an IDE. As the code takes shape, we discover which R packages it needs and add them to `generate-env.R`, rerunning `rix()` and rebuilding each time. In other words, following the same steps as described above. The specification grows from the bare environment (Listing 3) to one that covers the simulation (Listing 9):

Listing 9 Simulation environment, adding the packages the scripts need

```
rix(
  date = "2026-01-14",
  r_pkgs = c(
    "rix", "marginaleffects", "simhelpers", "rvinecopulib",
    "doParallel", "doRNG", "ggplot2", "cowplot", "dplyr", "svglite"
  ),
  ide = "rstudio",
  project_path = ".",
  overwrite = TRUE
)
```

Environment Specification

Declaring R package dependencies. The `r_pkgs` argument accepts a character vector listing all required R packages by their CRAN names. These packages will be installed from the version repository corresponding to the specified date or R version. It is important to list all packages that the analysis will load directly; dependencies of these packages are automatically resolved by Nix. For packages requiring specific versions not corresponding to the chosen date,

you are working in. We therefore recommend creating an **RStudio project file** (`.Rproj`) and opening that file when using Nix to ensure that RStudio is correctly associated with the intended project and environment.

researchers can specify exact versions using the syntax "packagename@version" (e.g., "ggplot2@2.2.1"). For packages available only on GitHub or other Git repositories, the git_pkgs argument accepts a list structure containing repository URLs and specific commit hashes. For example:

Listing 10 Example for the git_pkgs argument

```
git_pkgs = list(
  package_name = "marginaleffects",
  repo_url = "https://github.com/vincentarelbundock/marginaleffects",
  commit = "304bff91dc31ae28b227a8485bfa4f7bdc86d625"
)
```

This ensures that exact development versions are obtained (Rodrigues & Baumann, 2025). For our simulation study, all packages were used with their CRAN versions (see {rix} documentation for more details: https://docs.ropensci.org/rix/articles/installing-r-packages.html). Appendix B gives the reasons for adding each package in r_pkgs and tex_pkgs.

Again, users may add more packages "on-the-go" as one usually is not aware of all the packages that will be needed in advance. However, it is crucial that, after editing generate-env.R, we rebuild with nix-build and re-enter with nix-shell so the new packages become available.

Running the Simulation and Results

With the environment built, we run the simulation from inside the Nix shell so that it uses exactly the R version and packages declared in default.nix. This is how the results and plots reported in this manuscript were produced. For example, 03_run_simulation.R begins by loading the required packages and sourcing the other scripts it needs:

The simulation needs nothing beyond the environment built above (Listing 9). From within the Nix shell, the simulation runs (Listing 12) as follows:

In the same way, we run 04_performance_metrics.R (Listing 13), which loads the simulation results (in sim_results.rds) and calculates the performance metrics:

Listing 11 Code for running the simulation

```
library(marginaleffects)
...

# Source helper functions
source("Simulation/01_data_generation.R")
...
```

Listing 12 Running the complete simulation workflow

```
[nix-shell:~/Why-risk-it-when-you-can-rix-it]$
  Rscript Simulation/03_run_simulation.R
```

Similarly, `05_plots.R` (Listing 14) uses those saved metrics (in `performance_summary.rds`) to create the plots:

Because these results and plots come entirely from running the scripts inside the Nix environment, anyone can reproduce them. A reader downloads the project folder, builds the environment as in Steps 1-4 (`nix-build`, then `nix-shell`), and runs the same scripts in order. The advantage of executing within `nix-shell` is that all dependencies (the R version, packages, and system tools) match exactly those specified in `default.nix`. This approach does rely on running the scripts in sequence by hand. It guarantees a reproducible environment but does not formalize the workflow itself, leaving the dependencies between scripts implicit in the code rather than explicitly declared.

Step 6: Developing and Making the Complete Manuscript Reproducible[11]

To also build the manuscript, we extend the simulation environment (Listing 9) with the literate-programming tools it lacks: Quarto, {knitr}, and a set of LaTeX packages. We add these to `generate-env.R` and rebuild, giving the complete specification (Listing 15):

Including system-level dependencies. The `system_pkgs` parameter lists tools beyond R packages, such as Git, Pandoc, or, here, Quarto, which is a command-line tool rather than an R

[11] See the {rix} documentation for more: https://docs.ropensci.org/rix/articles/literate-programming.html

Listing 13 Running the performance metrics calculation

```
[nix-shell:~/Why-risk-it-when-you-can-rix-it]$
  Rscript Simulation/04_performance_metrics.R
```

Listing 14 Generating the visualization plots

```
[nix-shell:~/Why-risk-it-when-you-can-rix-it]$
  Rscript Simulation/05_plots.R
```

package. Not every system-level dependency needs listing. The {rvinecopulib} package, used in the simulation, links against C++ libraries such as Boost, Eigen, and RcppThread, yet Nix pulls these in automatically as dependencies of the package. Our manuscript also uses the `apaquarto` extension for APA formatting, stored in the project's `_extensions/` directory (Rodrigues & Baumann, 2025) (see {rix} documentation for more: https://docs.ropensci.org/rix/articles/installing-system-tools.html).

Specifying LaTeX packages. The `tex_pkgs` parameter lists the LaTeX packages needed for PDF compilation. When any are listed, Nix includes a minimal TeXLive distribution (`scheme-small`) and adds the requested packages on top. Finding the exact set often takes some trial and error, since Quarto's error messages during rendering indicate which packages are missing.

After rebuilding with `nix-build` and re-entering with `nix-shell`, Quarto is available, and we render the manuscript from the terminal (Listing 16):

This command executes all code chunks in the manuscript (i.e., loads the R packages, runs the data generation and estimation), calculates and incorporates results (e.g., Table 1) and figures (e.g., Figure 1), and generates a formatted PDF following APA style guidelines via the `apaquarto` extension (Schneider, 2024). This extension is saved in the project repo already. To add it to a project of your own, run the following inside the Nix shell, where Quarto is provided by the environment and therefore needs no separate installation (Listing 17):

Listing 15 Complete environment specification, adding the manuscript tools

```r
rix(
  date = "2026-01-14",
  r_pkgs = c(
    "rix", "quarto", "knitr", "marginaleffects",
    "simhelpers", "ggplot2", "doParallel", "doRNG", "cowplot",
    "dplyr", "svglite", "rvinecopulib"
  ),
  system_pkgs = c("quarto"),
  tex_pkgs = c("amsmath", "ninecolors", "apa7", "scalerel",
    "threeparttable", "threeparttablex", "endfloat", "environ",
    "multirow", "tcolorbox", "pdfcol", "tikzfill", "fontawesome5",
    "framed", "newtx", "fontaxes", "xstring", "wrapfig", "tabularray",
    "siunitx", "fvextra", "geometry", "setspace", "fancyvrb",
    "anyfontsize"
  ),
  ide = "rstudio",
  project_path = ".",
  overwrite = TRUE
)
```

Listing 16 Rendering the manuscript with Quarto

```
[nix-shell:~/Why-risk-it-when-you-can-rix-it]$
  quarto render Manuscript/article.qmd
```

The final document (.docx, .pdf, or .html) is saved directly in the project folder[12]. Because Quarto is installed as a system-level package in our Nix specification, the rendering occurs entirely within a fully reproducible environment, ensuring consistent output across machines regardless of local software configurations. If desired, the manuscript can also be reproduced interactively by opening the project folder in the user's preferred IDE.

As highlighted previously, we included the simulation code as code chunks for this manuscript. This was done simply for illustration purposes. In a real-world simulation study, users could simply run the simulation as in the previous sections and then load the needed results

[12] Although we use `apaquarto` in this example, many alternative manuscript templates are available, and Nix is agnostic to the specific template employed, provided the necessary extensions are installed.

Listing 17 Adding the apaquarto extension

```
[nix-shell:~/Why-risk-it-when-you-can-rix-it]$
  quarto add wjschne/apaquarto
```

(produced within the Nix shell) in the manuscript. However, the same remark holds with respect to the formalized workflow.

At this point, it is also worth noting that Nix shells do not fully isolate you from your existing system by default (as mentioned in footnote 7). For R users, this has a practical implication: packages installed in your regular R library (outside of Nix) could potentially be loaded when running R from within the Nix environment. The {rix} package addresses this automatically—when you call `rix()`, it also executes `rix_init()`, which creates a project-specific `.Rprofile`. This file configures R to ignore external package libraries and also disables `install.packages()` within the environment. The rationale is straightforward: any new packages should be added to `default.nix` and the environment rebuilt, preserving full reproducibility (Rodrigues & Baumann, 2025). However, for stricter isolation[13] that also prevents access to other system programs not specified in `default.nix`, use the `--pure` flag (Listing 18):

Listing 18 Activating the Nix environment with strict isolation

```
nix-shell --pure
```

Additional Considerations for Simulation Studies

In this section, we turn to considerations that matter especially for simulation studies but fall outside this tutorial's scope, since they depend on the details of each study. The first concerns random draws and reproducibility. The second concerns workflow orchestration and integration with tools researchers may already use. We do not cover these topics in depth. Our goal is to

[13] For example, when preparing this manuscript without the `--pure` flag, `quarto render` worked successfully. However, when using the `--pure` flag, the build failed. Running `quarto check` from within the Nix shell (i.e., `nix-shell --run "quarto check"`) revealed that Quarto was still accessing the system's LaTeX installation (`/Library/TeX/texbin`) rather than being restricted to only what was specified in `default.nix`.

clarify how they relate to {rix} and to point readers toward resources for more advanced use cases.

Reproducible Randomness and Parallel Execution for Data Generation

A simulation regenerates its data from a pseudo-random number generator, so its reproducibility depends on controlling that generator. The standard advice is to set the seed once and store it, so the same code reproduces the same sequence of draws (Morris et al., 2019). Simulation studies are typically organized as a set of conditions, each run for many repetitions, so the unit one wants to reproduce is a single repetition within a given condition. Storing the generator's state at the start of each condition-repetition cell, rather than only the initial seed, additionally allows any single cell to be reproduced in isolation, which is convenient for debugging (Morris et al., 2019). In R, the state lives in the variable `.Random.seed`, and such a procedure would look as follows (Listing 19):

With a sufficient number of repetitions the particular seed does not affect the conclusions. However, parallel processing complicates this (Morris et al., 2019; Siepe et al., 2024). In this case, a single seed may no longer be sufficient, because each worker would advance its own copy of the generator and could draw overlapping sequences. In practice, many researchers assign a distinct seed to each condition, for example `set.seed(123 + i)` for condition `i`, and let the repetitions within that condition draw sequentially from it. This keeps every condition reproducible and might not be a problem for most studies (given the properties of the default algorithm, i.e., Mersenne-Twister), but it does not by itself *guarantee* that the random sequences of different conditions never overlap. The recommended solution is to assign separate, non-overlapping random number streams (Morris et al., 2019; Siepe et al., 2024). R offers several packages for this (Gaujoux, 2025; Leydold, 2022; R Core Team, 2023), and we illustrate the approach with {doRNG}, which implements streams for {foreach} through the `%dorng%` operator (Listing 11). From a single master seed set before the loop, {doRNG} derives one independent stream per iteration in advance, before any work is dispatched. Each stream is bound to the iteration index rather than to a worker, so iteration seven always draws from stream seven no matter which core executes it. The number of cores (`ncore` in our script) changes only how

Listing 19 Storing and restoring the generator state to reproduce one condition-repetition cell

```r
# fix the starting position once
set.seed(123)

# 4 conditions, 1000 repetitions per condition
conditions <- expand.grid(n = c(50, 100), gamma2 = c(0, 0.3))
nsim <- 1000

# one slot per condition-repetition cell
states  <- matrix(list(), nrow(conditions), nsim)
results <- matrix(NA_real_, nrow(conditions), nsim)

# nested loops shown for illustration
for (cond in seq_len(nrow(conditions))) {
  for (r in seq_len(nsim)) {
    # state at the start of cell (cond, r)
    states[[cond, r]] <- .Random.seed
    # data generation advances the state
    x <- rnorm(conditions$n[cond])
    results[cond, r] <- mean(x)
  }
}

# later, reproduce repetition 3 of condition 2 on its own
.Random.seed <- states[[2, 3]]
identical(mean(rnorm(conditions$n[2])), results[2, 3])  # TRUE
```

iterations are scheduled in time, never which stream each one receives. A collaborator who reruns the study on four cores therefore reproduces data generated on twenty cores exactly. Moreover, because {doRNG} records the stream it assigned to each iteration in the `rng` attribute of the result, these states can be stored and later restored to reproduce the data of any single repetition within a condition in isolation, which the per-condition seed alone cannot do (Listing 20).

Because R's default generator can change across R versions, by pinning the R version {rix} keeps the generator and its algorithm fixed, so the same seed reproduces the same draws across machines and over time.

Listing 20 Reproducible parallel streams with {doRNG}

```r
library(doRNG)

# makes the streams below reproducible
set.seed(123)

# one slot per condition
rng_states <- vector("list", nrow(designs))

# nested loop over conditions
for (k in seq_len(nrow(designs))) {
  res <- foreach(
    i = seq_len(nsim),
    .packages = c("marginaleffects", "rvinecopulib")
  ) %dorng% {
    run_one_rep(designs$n[k], designs$gamma2[k])
  }
  # per-repetition stream states for this condition
  rng_states[[k]] <- attr(res, "rng")
  all_results[[k]] <- do.call(rbind, res)
}

# later, reproduce repetition 10 of condition 3 on its own
RNGkind("L'Ecuyer-CMRG")
.Random.seed <- rng_states[[3]][[10]]
rep10 <- run_one_rep(designs$n[3], designs$gamma2[3])
```

Workflow Orchestration: {targets} and {rixpress}

Complex simulation studies often benefit from workflow management systems that track dependencies between computational steps, cache intermediate results, and enable selective re-execution when inputs change. Two complementary approaches exist within the Nix ecosystem.

Using {targets} Within Nix. The {targets} package (Landau, 2021) provides workflow orchestration for R-based projects. To integrate {targets} with Nix, include `targets` in the `r_pkgs` parameter of `rix()` and execute the pipeline within `nix-shell` using `Rscript -e 'targets::tar_make()'`. A shell hook can also be added via the `shell_hook` argument to run the pipeline automatically when entering the Nix shell. This approach is ideal for projects that remain within R and do not require different environments for different pipeline steps (see {rix} documentation: https://docs.ropensci.org/rix/articles/reproducible-pipelines.html).

Using {rixpress} for Multi-Language Pipelines. The {rixpress} package (Rodrigues, 2025), a sister package to {rix}, uses Nix itself as the build automation tool rather than operating within a Nix environment. Each pipeline step becomes a Nix derivation, built in isolation with automatic caching based on content. The key advantage emerges in multi-language workflows: different steps can execute in different Nix-defined environments (e.g., one step using a specific version of R, another using Python, another using Julia). The interface, inspired by {targets}, uses functions like `rxp_r()`, `rxp_py()`, and `rxp_jl()` to define pipeline steps (see {rixpress} documentation: https://docs.ropensci.org/rixpress/articles/intro-concepts.html). The GitHub repository for this article directs interested readers to a demonstration of {rixpress} applied to this entire project.

Multi-Language Environment Support

While this tutorial focuses on R, researchers working across multiple programming languages can include Python or Julia in their environments. The `py_conf` parameter accepts a list specifying a Python version and required packages (Listing 21). Similarly, `jl_conf` enables Julia package installation. This capability is particularly useful, for example, for projects requiring statistical computing in R alongside machine learning pipelines in Python or numerical optimization in Julia (Rodrigues & Baumann, 2025) (see {rix} documentation for more: https://docs.ropensci.org/rix/articles/installing-r-packages.html).

Listing 21 Including Python packages

```
py_conf = list(py_version = "3.12", py_pkgs = c("polars", "pandas"))
```

Converting Existing {renv} Projects

Researchers with existing {renv} projects can migrate using the `renv2nix()` function, which reads an `renv.lock` file and generates an equivalent Nix expression. This is particularly valuable for projects where {renv} encountered system dependency issues or where stricter reproducibility guarantees are desired. Unlike {renv}, which captures R package versions but not the R interpreter or system libraries, Nix manages all layers of the software stack (see {rix} documentation: https://docs.ropensci.org/rix/articles/renv2nix.html).

Containerization with Docker

Nix and Docker are not necessarily mutually exclusive (Rodrigues & Baumann, 2026). Researchers already using Docker do not need to abandon it to benefit from Nix—the two can be combined by using Nix inside Docker containers to handle environment setup (Rodrigues & Baumann, 2025). This is particularly useful for deployment to cloud platforms or high-performance computing clusters where Docker is standard but Nix may not be available (see {rix} documentation: https://docs.ropensci.org/rix/articles/nix-inside-docker.html).

Discussion

Reproducibility in computational research is often treated as a matter of transparency—making data and code available. This tutorial has argued that transparency alone is insufficient without the ability to reliably reconstruct the computational environments in which analyses are executed. For simulation studies in particular, where results depend critically on software versions, system libraries, and random number generation, environment-level reproducibility is not optional but essential.

By introducing Nix and the {rix} package, we demonstrated a practical and accessible approach to fully specifying and rebuilding computational environments for simulation-based research. This approach enables analyses and manuscripts to be rerun identically across machines and over time, transforming reproducibility from an aspirational goal into a verifiable property of the research workflow.

Importantly, adopting environment reproducibility does not require abandoning existing analytic practices. Nix is agnostic to programming language, editor, workflow structure, and manuscript template, allowing researchers to retain familiar tools while strengthening the reliability of their work. In this sense, reproducible environments serve as enabling infrastructure—supporting, rather than replacing, other best practices such as version control, workflow orchestration, and transparent reporting.

If reproducibility is to function as a cornerstone of cumulative science, then the ability to reconstruct computational environments must become a routine part of methodological practice. Tools such as Nix and {rix} lower the barrier to achieving this goal, making fully reproducible simulation research feasible without requiring deep systems expertise. We hope this tutorial helps normalize environment-level reproducibility as a standard component of rigorous computational research in psychology and beyond.

Glossary

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Table 1

Performance metrics for ACE estimator across simulation conditions. Values in parentheses are Monte Carlo standard errors (MCSE).

Sample Size	Confounding	Relative Bias	Relative RMSE	Coverage	CI Width
50	none	0.957 (0.042)	0.422 (0.035)	0.910 (0.029)	0.224 (0.004)
100	none	1.013 (0.026)	0.256 (0.020)	0.930 (0.026)	0.159 (0.002)
2000	none	0.997 (0.006)	0.059 (0.005)	0.960 (0.020)	0.036 (0.000)
50	mild	1.173 (0.036)	0.395 (0.034)	0.890 (0.031)	0.222 (0.004)
100	mild	1.157 (0.028)	0.318 (0.026)	0.870 (0.034)	0.154 (0.002)
2000	mild	1.114 (0.006)	0.130 (0.006)	0.510 (0.050)	0.035 (0.000)
50	severe	1.200 (0.044)	0.482 (0.054)	0.870 (0.034)	0.223 (0.005)
100	severe	1.235 (0.030)	0.379 (0.027)	0.830 (0.038)	0.157 (0.002)
2000	severe	1.257 (0.007)	0.266 (0.007)	0.020 (0.014)	0.035 (0.000)

Table 2

Glossary of computing terms used in the main text.

Term	Definition
R interpreter	The program that runs R code. Two machines can share an R version yet differ in how the interpreter was built, which can change results.
System-level libraries	Shared operating-system software that R packages depend on (e.g., BLAS/LAPACK, C++ backends); not installed by `install.packages()`.
Workflow orchestration	Automating the order of analysis steps and re-running only those affected by a change (e.g., the {targets} package).
Containerization	Bundling software with its dependencies into an isolated, portable unit (e.g., Docker); does not by itself guarantee identical future rebuilds.
Mutable upstream repositories	Online software sources that can change over time, so a later rebuild may pull different versions.
Functional programming language	A language whose functions depend only on their inputs; Nix is one, but {rix} writes it for you.

Figure 1

The computational environment as a set of nested layers, based on Peikert & Brandmaier (2021).

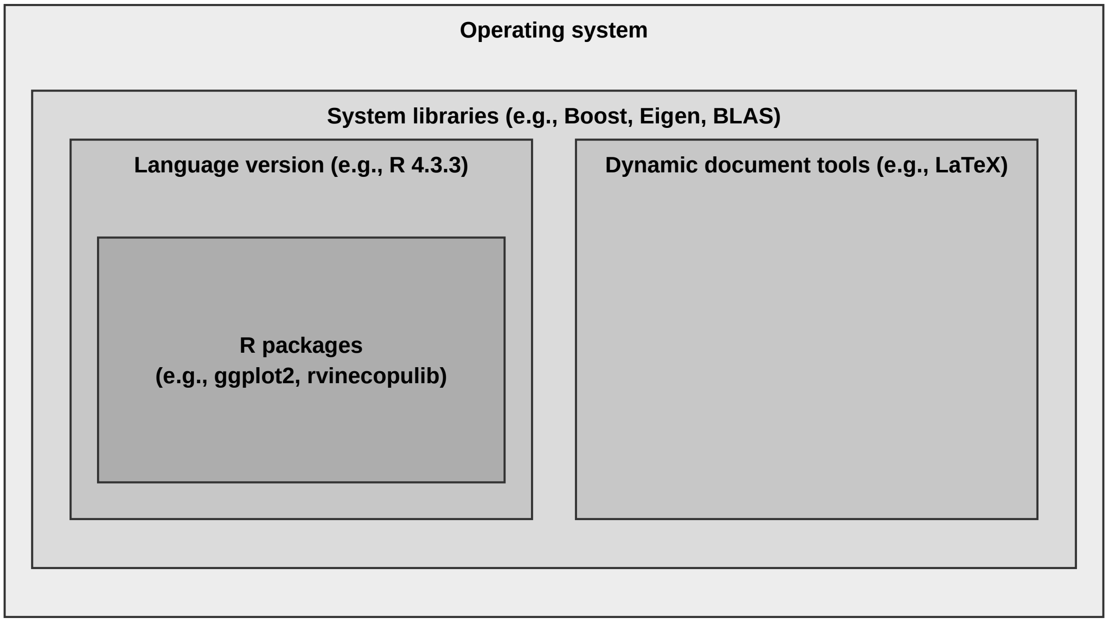
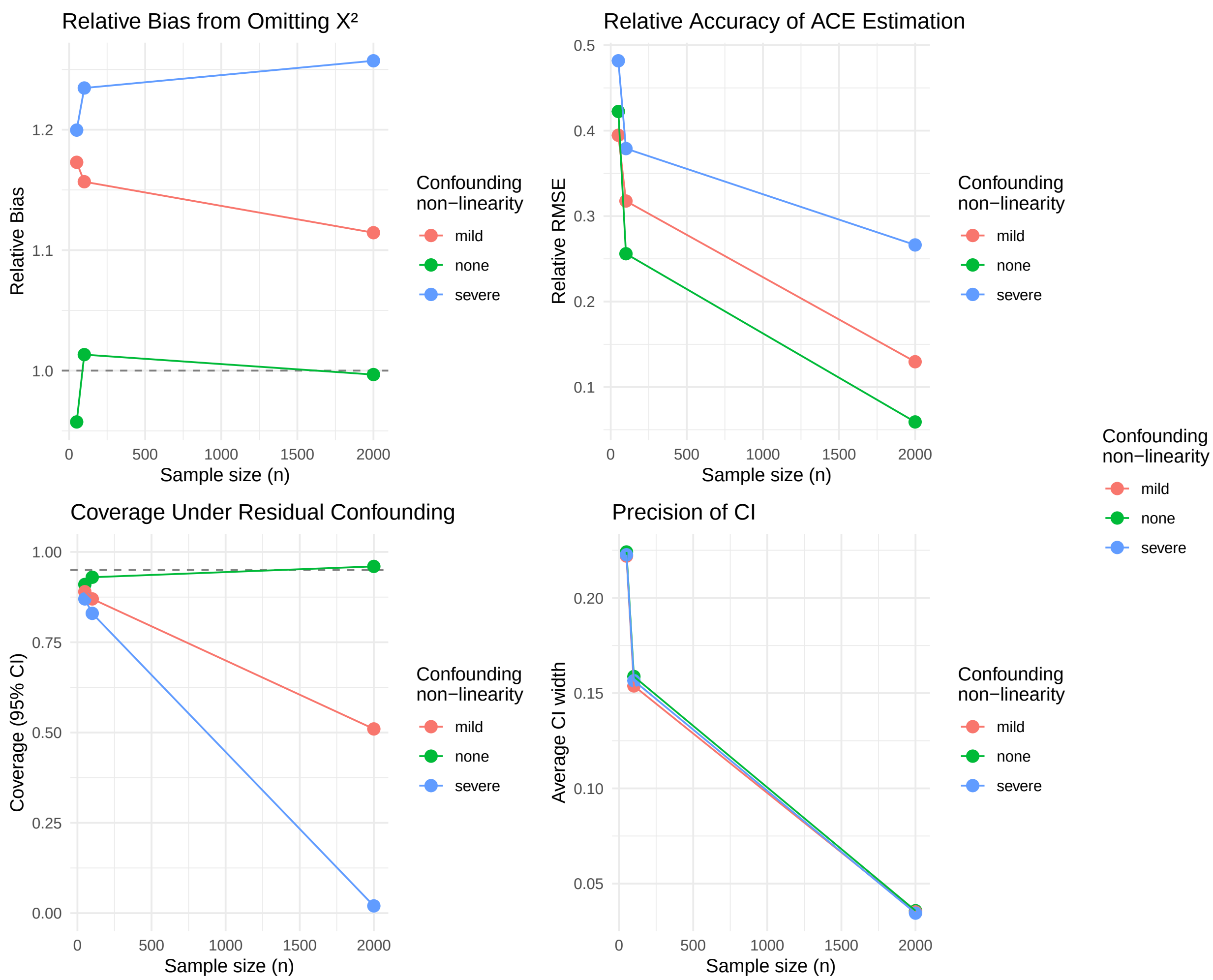


Figure 2

Performance of ACE estimator across sample sizes and confounding severity. Panel A shows relative bias, Panel B shows relative RMSE, Panel C shows coverage probability of 95% confidence intervals (dashed line at nominal 0.95 level), and Panel D shows average confidence interval width. Results demonstrate that model misspecification induces systematic bias that persists across sample sizes, while increasing sample size improves precision but not accuracy under misspecification.



Appendix A

Simulation Study Design

Here we present a rather short description following recommendations from previous research, but ideally even more may be reported (Morris et al., 2019; Pawel et al., 2025; Siepe et al., 2024; White et al., 2024). This mimics a methods or similar section in articles.

Factorial Design. The simulation employs a full factorial design with two factors: sample size ($n \in \{50, 100, 2000\}$) and degree of confounding non-linearity ($\gamma_2 \in \{0, 0.3, 0.8\}$, labeled as none, mild, and severe). The parameter γ_2 controls the strength of the quadratic confounder effect on the outcome (see Data Generation). This yields nine conditions, each replicated $K = 100$ times.

Data Generation. For each replication, data are generated following a causal structure where a confounder X_2 affects both treatment assignment and the outcome. The confounder and treatment error term are generated using the {rvinecopulib} package: pairs (U_1, U_2) are drawn from an independence copula via `rbicop()`, then transformed to standard normals via $X_2 = \Phi^{-1}(U_1)$ and $\epsilon = \Phi^{-1}(U_2)$. The independence copula is simply $C(u, v) = uv$, meaning the resulting uniforms are independent—mathematically equivalent to calling `rnorm()` directly. We use {rvinecopulib} intentionally because it depends on C++ libraries.

Treatment assignment follows $X_1 = \alpha_0 + \alpha_1 X_2 + \alpha_2 X_2^2 + \epsilon$ where $\alpha_0 = 0$, $\alpha_1 = 0.5$, and $\alpha_2 = 0.2$. This creates confounding because X_2 influences treatment assignment through both linear and quadratic terms. The binary outcome is generated from the true logistic regression model:

$$\text{logit}(P(Y = 1 \mid X_1, X_2)) = \beta_0 + \beta_1 X_1 + \gamma_1 X_2 + \gamma_2 X_2^2$$

with parameters $\beta_0 = -0.5$, $\beta_1 = 0.7$ (the causal effect of interest), $\gamma_1 = -0.4$, and γ_2 varying by condition. The analyst misspecifies the outcome model by omitting the quadratic confounder term, fitting instead:

$$\text{logit}(P(Y = 1 \mid X_1, X_2)) = \beta_0 + \beta_1 X_1 + \gamma_1 X_2$$

This misspecification creates residual confounding because the omitted term $\gamma_2 X_2^2$ is correlated with X_1 (since X_1 depends on both X_2 and X_2^2), violating the conditional exchangeability assumption given linear adjustment alone.

Estimand. The target estimand is the average causal effect (ACE) of X_1 on Y, properly adjusted for confounding:

$$\text{ACE}(X_1) = \mathbb{E}\left[\frac{\partial P(Y = 1 \mid X_1, X_2)}{\partial X_1}\right] = \mathbb{E}\left[\beta_1 \cdot \frac{\exp(\eta)}{(1 + \exp(\eta))^2}\right]$$

where $\eta = \beta_0 + \beta_1 X_1 + \gamma_1 X_2 + \gamma_2 X_2^2$ is the correctly specified linear predictor, and the expectation is taken over the joint distribution of (X_1, X_2). For each γ_2 condition, the "true" ACE (denoted θ) is approximated once using a very large sample ($N = 200,000$) with the correctly specified model including X_2^2.

Estimator. The causal effect is estimated from the misspecified model (omitting X_2^2) as:

$$\widehat{\text{ACE}}(X_1) = \frac{1}{n}\sum_{i=1}^{n} \tilde{\beta}_1 \cdot \frac{\exp(\tilde{\eta}_i)}{(1 + \exp(\tilde{\eta}_i))^2}$$

where $\tilde{\eta}_i = \tilde{\beta}_0 + \tilde{\beta}_1 X_{i1} + \tilde{\gamma}_1 X_{i2}$ and $\tilde{\boldsymbol{\beta}} = (\tilde{\beta}_0, \tilde{\beta}_1, \tilde{\gamma}_1)$ are maximum likelihood estimates from the misspecified logistic regression.

Performance Criteria. Table B1 presents the performance criteria used to evaluate the ACE estimator across simulation conditions.

Computational Details. The simulation was conducted on a MacBook Pro (Apple M4 Pro Chip), running macOS Sequoia 15.7.3. All analyses were performed in R (version 4.5.2). Parallel processing was implemented through the {doParallel} package [version 1.0.17; Corporation and Weston (2022)], with {doRNG} [version 1.8.6.2; Gaujoux (2025)] to ensure independent and reproducible random number streams. For data generation we used the {rvinecopulib} package [version 0.7.3.1.0; Nagler and Vatter (2025)]. The estimator was implemented using the {marginaleffects} package [version 0.31.0; Arel-Bundock et al. (2024)]. Data wrangling was performed with {dplyr} [version 1.1.4; Wickham et al. (2023)]. Method performance was assessed through multiple metrics following the formulas from the {simhelpers} package [version 0.3.1; Joshi and Pustejovsky (2025)]. Figures were produced with {ggplot2} [version 4.0.1; Wickham (2016)] and {cowplot} [version 1.2.0; Wilke (2025)].

Appendix B

Clarifying the packages used

R Packages

Reproducibility Infrastructure

rix: Generates the Nix expression (`default.nix`) for reproducible environments.

Simulation Study

These packages are used in the `Simulation_Scripts/` folder:

rvinecopulib (used in `01_data_generation.R`): Generates correlated data via copulas using the `rbicop()` function.

marginaleffects (used in `02_models.R`): Computes average causal effects via the `avg_slopes()` function.

doParallel (used in `03_run_simulation.R`): Enables parallel `foreach` loops across CPU cores.

doRNG (used in `03_run_simulation.R`): Makes parallel RNG reproducible.

simhelpers (used in `04_performance_metrics.R`): Calculates bias, RMSE, and coverage metrics.

ggplot2 (used in `05_plots.R`): Creates simulation result visualizations.

cowplot (used in `05_plots.R`): Combines plots with `plot_grid()` and extracts legends.

dplyr (used in `article.qmd`): Data wrangling for results reported in Table 1.

Dynamic Document Generation

quarto: R interface to invoke Quarto rendering.

knitr: Executes R code chunks in `.qmd` files.

svglite: SVG graphics device; apaquarto sets `dev: svglite` for HTML output.

LaTeX Packages

Required by apaquarto Extension

These are loaded in apaquarto's `header.tex` or `apatemplate.tex`:

amsmath: Math environments (`align`, `equation`, etc.).

threeparttablex: Tables with notes below (APA table format).

tcolorbox: Callout boxes (note, warning, tip blocks).

fontawesome5: Icons in callouts.

multirow: Table cells spanning multiple rows.

newtx: Times-like fonts (default when no custom `mainfont`).

geometry: Page margins.

Dependencies of apaquarto Packages

environ: Dependency of tcolorbox.

pdfcol: Dependency of tcolorbox.

tikzfill: Dependency of tcolorbox.

fontaxes: Dependency of newtx.

xstring: Template conditionals.

scalerel: Dependency of apa7 class.

Required by apa7 Document Class

apa7: The document class itself (`\documentclass{apa7}`).

endfloat: Moves floats to end of document in manuscript mode.

threeparttable: Dependency for threeparttablex.

Required by Quarto PDF Rendering

framed: Shaded/framed regions for callouts.

fvextra: Enhanced verbatim for syntax-highlighted code blocks.

fancyvrb: Verbatim environments for code display.

setspace: Line spacing; also used in `article.qmd` for single-spaced code blocks.

anyfontsize: Arbitrary font sizes in code blocks.

Additional Packages

ninecolors: Extended color palettes.

wrapfig: Text wrapping around figures.

tabularray: Modern table typesetting.

siunitx: SI units and number formatting.

Table B1

Performance criteria for evaluating the ACE estimator. $\hat{\theta}_k$ denotes the ACE estimate from replication k (for $k = 1, \ldots, K$), where $K = 1000$ is the number of replications, and θ denotes the true ACE for a given condition. For coverage and width criteria, A_k and B_k denote the lower and upper endpoints of the 95% confidence interval from replication k, $W_k = B_k - A_k$ is the interval width, c_β is the estimated coverage probability, and $I(\cdot)$ is an indicator function equaling 1 if the condition is true and 0 otherwise. The Monte Carlo standard error (MCSE) quantifies the simulation uncertainty in each performance measure estimate

Criterion	Estimate	MCSE
Bias	$\frac{1}{K}\sum_{k=1}^{K}\hat{\theta}_k - \theta$	$\sqrt{\frac{1}{K(K-1)}\sum_{k=1}^{K}(\hat{\theta}_k - \bar{\hat{\theta}})^2}$
Variance	$\frac{1}{K-1}\sum_{k=1}^{K}(\hat{\theta}_k - \bar{\hat{\theta}})^2$	$\sqrt{\frac{K-1}{K}} \cdot \frac{1}{K-1}\sum_{k=1}^{K}(\hat{\theta}_k - \bar{\hat{\theta}})^2$
RMSE	$\sqrt{\frac{1}{K}\sum_{k=1}^{K}(\hat{\theta}_k - \theta)^2}$	$\sqrt{\frac{K-1}{K}\sum_{j=1}^{K}\left(\sqrt{(\hat{\theta}_j - \theta)^2} - \overline{RMSE}\right)^2}$
Relative Bias	$\frac{1}{\theta K}\sum_{k=1}^{K}\hat{\theta}_k$	$\frac{1}{\theta}\sqrt{\frac{1}{K(K-1)}\sum_{k=1}^{K}(\hat{\theta}_k - \bar{\hat{\theta}})^2}$
Relative RMSE	$\frac{1}{\theta}\sqrt{\frac{1}{K}\sum_{k=1}^{K}(\hat{\theta}_k - \theta)^2}$	$\frac{1}{\theta}\sqrt{\frac{K-1}{K}\sum_{j=1}^{K}\left(\sqrt{(\hat{\theta}_j - \theta)^2} - \overline{RMSE}\right)^2}$
Coverage	$\frac{1}{K}\sum_{k=1}^{K}I(A_k \leq \theta \leq B_k)$	$\sqrt{\frac{c_\beta(1-c_\beta)}{K}}$
Width	$\frac{1}{K}\sum_{k=1}^{K}(B_k - A_k)$	$\sqrt{\frac{1}{K(K-1)}\sum_{k=1}^{K}(W_k - \bar{W})^2}$